**MATH 6660 — Homework 1**

Hasan

# Problem 1(a)

**Given.** $\mathcal{L}(u) := -\big(a(x)u'\big)'$ on $0 < x < 1$ with

$$0 < a_1 \le a(x) \le a_2 < \infty, \tag{1}$$

so in particular $a$ is real-valued. For any $u, v$ vanishing at both endpoints, Green's formula holds:

$$\int_0^1 \big[\, u\,\mathcal{L}(v) - v\,\mathcal{L}(u)\,\big]\,\mathrm{d}x = 0. \tag{2}$$

**Problem.** For the eigenvalue problem

$$\mathcal{L}(\phi) = \lambda\,\phi, \qquad \phi(0) = \phi(1) = 0, \qquad \phi \not\equiv 0, \tag{3}$$

prove that $\lambda$ must be real. ($\overline{\cdot}$ denotes complex conjugation.)

*Note.* Neither $\phi(0) = \phi(1) = 0$ nor $\phi \not\equiv 0$ is printed in the assignment. The boundary conditions come from the word *associated* — the eigenvalue problem is associated with the boundary value problem above, whose solutions satisfy $u(0) = u(1) = 0$, so its eigenfunctions live in the same space; and $\phi \not\equiv 0$ is part of the definition of *eigenfunction*, exactly as the zero vector is never an eigenvector. Both are needed: without the boundary conditions every $\lambda \in \mathbb{C}$ would be an eigenvalue (for $a \equiv 1$, $\phi = e^{\mathrm{i}\sqrt{\lambda}x}$ solves $-\phi'' = \lambda\phi$), and without $\phi \not\equiv 0$ step 3 below reads $0 = (\overline{\lambda} - \lambda)\cdot 0$, which holds for every $\lambda$.

## Solution

**Step 1: $(\overline{\lambda}, \overline{\phi})$ is also an eigenpair.** Conjugate (3). Conjugation commutes with $\mathrm{d}/\mathrm{d}x$, and $a$ is real by (1), so $\overline{\big(-(a\phi')'\big)} = -\big(a\,(\overline{\phi})'\big)' = \mathcal{L}(\overline{\phi})$. Hence

$$\mathcal{L}(\overline{\phi}) = \overline{\lambda}\,\overline{\phi}, \tag{4}$$

and $\overline{\phi}(0) = \overline{\phi}(1) = 0$. *(Realness of $a$ is used only here; for complex $a$ this step fails.)*

**Step 2: apply (2) with $u = \phi$, $v = \overline{\phi}$.** Both vanish at the endpoints, so

$$0 = \int_0^1 \big[\, \phi\,\mathcal{L}(\overline{\phi}) - \overline{\phi}\,\mathcal{L}(\phi)\,\big]\,\mathrm{d}x = \int_0^1 \big[\overline{\lambda}\,\phi\overline{\phi} - \lambda\,\overline{\phi}\phi\big]\,\mathrm{d}x = \big(\overline{\lambda} - \lambda\big)\int_0^1 |\phi|^2\,\mathrm{d}x, \tag{5}$$

using (4), (3), and $\phi\overline{\phi} = |\phi|^2$.

**Step 3: conclude.** $\phi$ is continuous and $\phi \not\equiv 0$, so $\int_0^1 |\phi|^2\,\mathrm{d}x > 0$. Dividing (5) by it gives $\overline{\lambda} = \lambda$, i.e.

$$\boxed{\lambda \in \mathbb{R}} \tag{6}$$

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## Problem 1(b)

**Problem.** Let $\phi_1, \phi_2$ be eigenfunctions of (3) corresponding to eigenvalues $\lambda_1 \neq \lambda_2$. Prove they are *orthogonal*:

$$\int_0^1 \phi_1(x)\,\phi_2(x)\,\mathrm{d}x = 0. \tag{7}$$

### Solution

By part (a) the eigenvalues are real, so the eigenfunctions may be taken real-valued and (2) applies to them directly. Both $\phi_1$ and $\phi_2$ vanish at $x = 0$ and $x = 1$, so (2) with $u = \phi_1$, $v = \phi_2$ gives

$$0 = \int_0^1 \big[\,\phi_1\,\mathcal{L}(\phi_2) - \phi_2\,\mathcal{L}(\phi_1)\,\big]\,\mathrm{d}x = \int_0^1 \big[\lambda_2\,\phi_1\phi_2 - \lambda_1\,\phi_2\phi_1\big]\,\mathrm{d}x = \big(\lambda_2 - \lambda_1\big)\int_0^1 \phi_1\phi_2\,\mathrm{d}x, \tag{8}$$

using $\mathcal{L}(\phi_i) = \lambda_i\phi_i$. Since $\lambda_1 \neq \lambda_2$ the factor $\lambda_2 - \lambda_1$ is nonzero, so dividing (8) by it gives (7). ■

## Problem 1(c)

**Problem.** Prove that all eigenvalues of (3) are strictly positive: $\lambda > 0$.

*Note.* This is the first part in which the lower bound $a \geq a_1 > 0$ of (1) is actually used; parts (a) and (b) needed only that $a$ is real-valued. It is what rules out $\lambda = 0$.

### Solution

Green's formula (2) is vacuous here, since $u = v = \phi$ makes it read $0 = 0$; instead integrate by parts once. Multiplying (3) by $\phi$ and integrating,

$$\int_0^1 \phi\, \mathcal{L}(\phi)\, \mathrm{d}x = \lambda \int_0^1 \phi^2\, \mathrm{d}x. \tag{9}$$

The boundary term vanishes because $\phi(0) = \phi(1) = 0$, so the left-hand side is

$$\int_0^1 \phi \left(-(a\phi')'\right) \mathrm{d}x = -\Big[\, a\,\phi'\phi \,\Big]_0^1 + \int_0^1 a\,(\phi')^2\, \mathrm{d}x = \int_0^1 a\,(\phi')^2\, \mathrm{d}x. \tag{10}$$

Combining (9) and (10),

$$\lambda \int_0^1 \phi^2\, \mathrm{d}x = \int_0^1 a\,(\phi')^2\, \mathrm{d}x \qquad \Longrightarrow \qquad \lambda = \frac{\displaystyle\int_0^1 a\,(\phi')^2\, \mathrm{d}x}{\displaystyle\int_0^1 \phi^2\, \mathrm{d}x}\,. \tag{11}$$

**$\lambda \geq 0$.** The denominator is positive since $\phi$ is continuous and $\phi \not\equiv 0$; the numerator is non-negative since $a \geq a_1 > 0$ by (1) and $(\phi')^2 \geq 0$.

**$\lambda \neq 0$.** Suppose $\lambda = 0$. Then $\int_0^1 a(\phi')^2\, \mathrm{d}x = 0$, and since $a \geq a_1 > 0$,

$$0 = \int_0^1 a\,(\phi')^2\, \mathrm{d}x \;\geq\; a_1 \int_0^1 (\phi')^2\, \mathrm{d}x \;\geq\; 0 \qquad \Longrightarrow \qquad \int_0^1 (\phi')^2\, \mathrm{d}x = 0. \tag{12}$$

As $\phi'$ is continuous this forces $\phi' \equiv 0$, so $\phi$ is constant; with $\phi(0) = 0$ that gives $\phi \equiv 0$, contradicting $\phi \not\equiv 0$. Hence

$$\boxed{\lambda > 0} \tag{13}$$

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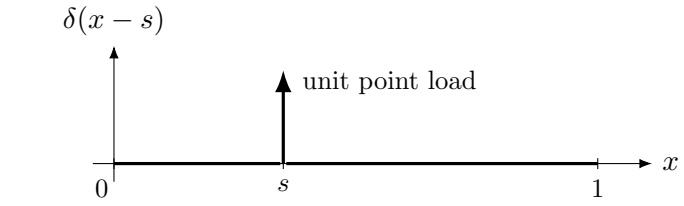
## Problem 1(d)

**Problem.** Let $a(x) \equiv 1$, so $\mathcal{L} = -\mathrm{d}^2/\mathrm{d}x^2$, and take the load to be a unit point load at $x = s$:

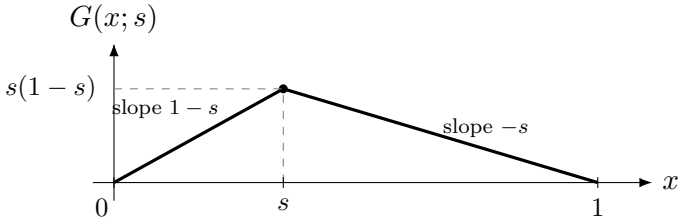
$$\mathcal{L}(G) = -G'' = \delta(x - s), \qquad G(0; s) = G(1; s) = 0, \qquad 0 < s < 1. \tag{14}$$

Compute $G(x; s)$ by hand. (In $G(x; s)$, $x$ is the observation point and $s$ the location of the load.)

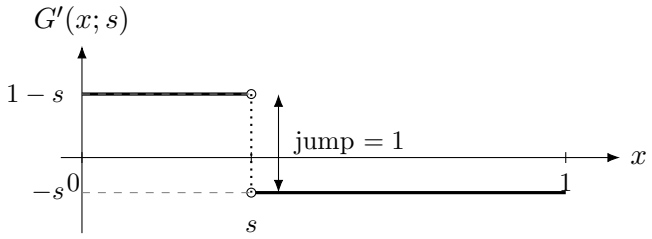
*Note.* "Delta" and "Dirac delta" are the same object; *Dirac* is just the full name. But $\delta$ does four different jobs in this course, told apart by what follows it: $\delta(x-s)$, with an argument, is the *Dirac delta* — a distribution, not a function: it has no value at $x = s$, only the property $\int \delta(x - s)\,\mathrm{d}x = 1$; $\delta_{ij}$, with subscripts, is the *Kronecker delta*, the number 1 if $i = j$ and 0 otherwise; a bare $\delta > 0$ is a small positive real; and $\delta u$ denotes a *variation* (a virtual displacement).



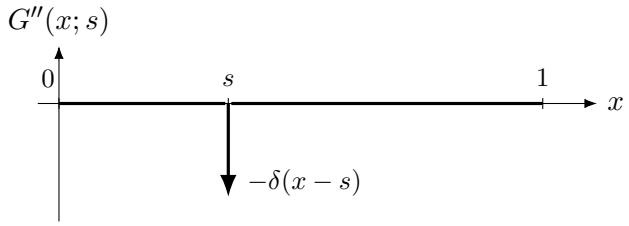

**the load**: area 1 concentrated at $x = s$




**the response**: continuous, with a *kink* under the load




**the slope**: piecewise constant, with a *jump* of exactly 1 at $x = s$




**the second derivative**: $G'' = -\delta(x - s)$ — the load recovered, with the sign of (14)

Each panel has its own vertical axis, named on it: the load $\delta(x - s)$, the response $G(x; s)$, its slope $G'(x; s)$, and its curvature $G''(x; s)$. The horizontal axis is $x$ in all four. Reading downwards is differentiating, and the last panel returns to the first up to sign: $G'' = -\delta$, which is (14).

## Solution

**Piecewise form.** For $x \neq s$ the right-hand side of (14) vanishes, so $G'' = 0$ and $G$ is linear on each side. The boundary conditions remove one constant from each piece:

$$G(x;s) = \begin{cases} A\,x, & 0 \le x \le s, \\ C\,(x-1), & s \le x \le 1. \end{cases} \tag{15}$$

**Matching at $x = s$.** Two conditions determine $A$ and $C$.
*Continuity.* $G$ must be continuous: a jump in $G$ would put a $\delta'$ into $G''$, which (14) does not have. Hence

$$A\,s = C\,(s-1). \tag{16}$$

*Slope jump.* From (15) the slope is constant on each side:

$$G'(x) = A \quad \text{on } (0,s), \qquad G'(x) = C \quad \text{on } (s,1). \tag{17}$$

Now integrate (14) across $x = s$, over $[s-\epsilon, s+\epsilon]$. Taking the **left-hand side** one step at a time:

$$\begin{aligned} \int_{s-\epsilon}^{s+\epsilon} (-G'')\,\mathrm{d}x &= -\int_{s-\epsilon}^{s+\epsilon} G''\,\mathrm{d}x && \text{pull the minus sign out} \\ &= -\Big[\,G'\,\Big]_{s-\epsilon}^{s+\epsilon} && G'' = (G')', \text{ so the fundamental theorem applies} \\ &= -\Big[\,G'(s+\epsilon) - G'(s-\epsilon)\,\Big] && \text{evaluate at the two endpoints} \\ &= G'(s-\epsilon) - G'(s+\epsilon) && \text{distribute the minus sign} \\ &= A - C && \text{by (17): left slope } A\text{, right slope } C. \end{aligned}$$

The **right-hand side** is 1, because the spike sits inside the interval and $\delta$ has unit area:

$$\int_{s-\epsilon}^{s+\epsilon} \delta(x-s)\,\mathrm{d}x = 1.$$

Equating the two sides,

$$A - C = 1. \tag{18}$$

Note the answer does not depend on $\epsilon$, so no limit is needed.
**Solve.** Substituting $A = C + 1$ from (18) into (16) gives $(C+1)s = Cs - C$, so $C = -s$ and $A = 1 - s$. Therefore

$$\boxed{G(x;s) = \begin{cases} x\,(1-s), & 0 \le x \le s, \\ s\,(1-x), & s \le x \le 1, \end{cases} \qquad \text{i.e.} \qquad G(x;s) = \min(x,s)\,\big[1 - \max(x,s)\big].} \tag{19}$$

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**Check.** $G(0;s) = G(1;s) = 0$; at $x = s$ both branches give $s(1-s)$, so $G$ is continuous; and $G'(s^-) - G'(s^+) = (1-s) - (-s) = 1$, as required by (18).

*Note.* The point load produces a *kink* in $G$, not a jump: $G$ is continuous while $G'$ jumps by 1. This is why $\int_0^1 (G')^2\,\mathrm{d}x$ is finite but $\int_0^1 (G'')^2\,\mathrm{d}x$ is not.

## Problem 1(e)

**Problem.** Let $a(x) \equiv 1$. Show that $G(x;s) = G(s;x)$, and, using an appropriate application of Green's formula, show that for a general load $f \in C^0(0,1)$ the solution of

$$\mathcal{L}(u) = -u'' = f(x), \qquad u(0) = u(1) = 0 \tag{20}$$

is given by

$$u(x) = \int_0^1 f(s)\, G(s;x)\, \mathrm{d}s. \tag{21}$$

### Solution

**(i) Symmetry.** From the compact form in (19), $G(x;s) = \min(x,s)\big[1 - \max(x,s)\big]$. Interchanging $x$ and $s$ changes neither the smaller nor the larger of the two, so

$$G(s;x) = \min(s,x)\big[1 - \max(s,x)\big] = \min(x,s)\big[1 - \max(x,s)\big] = G(x;s). \tag{22}$$

■

**(ii) The representation formula.** Fix $x \in (0,1)$ and put

$$\varphi(s) := G(s;x), \tag{23}$$

a function of $s$ with $x$ as a parameter. By (14), with the roles of the two arguments renamed,

$$\mathcal{L}(\varphi)(s) = -\varphi''(s) = \delta(s-x), \qquad \varphi(0) = \varphi(1) = 0. \tag{24}$$

Both $u$ and $\varphi$ vanish at $s = 0$ and $s = 1$, so Green's formula (2) applies to the pair $(u, \varphi)$ in the variable $s$:

$$\begin{aligned} 0 &= \int_0^1 \Big[\, u(s)\, \mathcal{L}(\varphi)(s) - \varphi(s)\, \mathcal{L}(u)(s) \,\Big] \mathrm{d}s && \text{Green's formula (2)} \\ &= \int_0^1 \Big[\, u(s)\, \delta(s-x) - G(s;x)\, f(s) \,\Big] \mathrm{d}s && \text{by (24) and (20)} \\ &= u(x) - \int_0^1 f(s)\, G(s;x)\, \mathrm{d}s && \text{sifting: } \textstyle\int u(s)\delta(s-x)\,\mathrm{d}s = u(x). \end{aligned} \tag{25}$$

Rearranging (25) gives (21), and by (22) it may equally be written $u(x) = \int_0^1 f(s)\, G(x;s)\, \mathrm{d}s$. ■

*Note.* $\varphi = G(\cdot\,; x)$ is not classically $C^2$ — its second derivative is a delta — so (25) is a *formal* application of (2). This is standard for Green's-function derivations: the identity is justified by splitting the integral at $s = x$ and integrating by parts on each side, where $\varphi$ is smooth, the kink contributing exactly the jump that produces $u(x)$.

## Problem 2

**Problem.** With the scalar product and induced norm

$$(v,\, w) := \int_0^1 v(x)\, w(x)\, \mathrm{d}x \qquad \text{and} \qquad \|v\| := \sqrt{(v,\, v)}\,, \tag{26}$$

prove *Cauchy's inequality*

$$\big|\,(v,\, w)\,\big| \;\le\; \|v\|\,\|w\|. \tag{27}$$

*Note.* This problem is self-contained: no operator $\mathcal{L}$, no boundary value problem, no boundary conditions. Here $v$ and $w$ are any two continuous functions on $[0,1]$.

### Solution

**Step 1: $P$ is a quadratic in $t$.** Following the hint, set

$$P(t) := \|v - tw\|^2 = (v - tw,\, v - tw), \qquad t \in \mathbb{R}. \tag{28}$$

The scalar product (26) is bilinear and symmetric, since $(v,w) = \int_0^1 vw\,\mathrm{d}x = \int_0^1 wv\,\mathrm{d}x = (w,v)$. Expanding,

$$P(t) = (v,v) - 2t\,(v,w) + t^2(w,w) = \underbrace{\|w\|^2}_{A}\, t^2 \;\underbrace{-\,2(v,w)}_{B}\, t \;+\; \underbrace{\|v\|^2}_{C}\,. \tag{29}$$

**Step 2: $P$ is never negative.** Directly from (26),

$$P(t) = \int_0^1 \big(v(x) - t\,w(x)\big)^2\,\mathrm{d}x \;\ge\; 0 \qquad \text{for every } t \in \mathbb{R}, \tag{30}$$

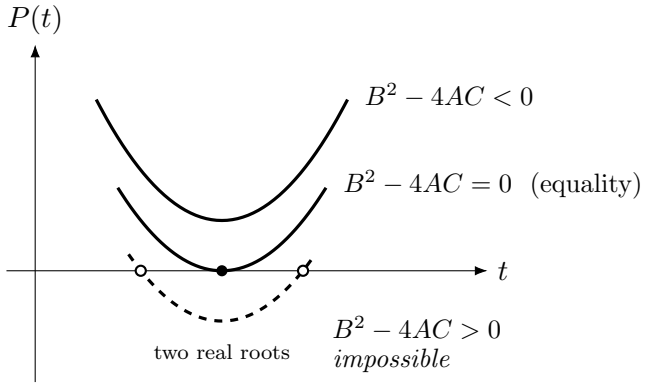
because the integrand is a square.

**Step 3: the degenerate case.** If $w \equiv 0$ then $\|w\| = 0$ and $(v,w) = 0$, so (27) reads $0 \le 0$ and holds. This case must be taken separately, since then $A = 0$ and $P$ is not quadratic.

**Step 4: the discriminant.** Assume $w \not\equiv 0$. Then $A = \|w\|^2 = \int_0^1 w^2\,\mathrm{d}x > 0$, because $w$ is continuous and not identically zero, so $P$ is a genuine upward parabola. A parabola opening upwards that never dips below zero meets the axis at most once, so its discriminant is non-positive:

$$B^2 - 4AC \le 0 \quad\Longrightarrow\quad 4(v,w)^2 - 4\|w\|^2\|v\|^2 \le 0 \quad\Longrightarrow\quad (v,w)^2 \le \|v\|^2\|w\|^2. \tag{31}$$

Both sides of the last inequality are non-negative, so taking square roots gives (27). ∎




$P(t) = \int_0^1 (v - tw)^2\,\mathrm{d}x$ is a sum of squares, so its graph can never dip below the $t$-axis. That rules out the dashed curve, which would have two real roots, and therefore forces $B^2 - 4AC \le 0$ — exactly (27). The borderline case is tangency (solid dot), at $t^* = (v,w)/\|w\|^2$, where equality holds and $v = t^*w$.

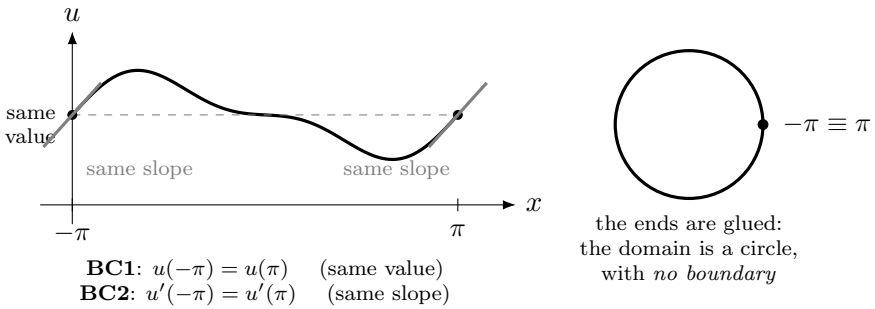
*Note.* Equality holds exactly when $P$ *does* have a real root, i.e. when $v = tw$ for some $t$ — the two functions parallel. For vectors this is the case $|\cos\theta| = 1$.

## Problem 3(a)

**Problem.** Recast the periodic boundary value problem

$$\textbf{ODE}: \ -u'' + q(x)\,u = f(x), \qquad \textbf{BC1}: \ u(-\pi) = u(\pi), \qquad \textbf{BC2}: \ u'(-\pi) = u'(\pi), \tag{32}$$

with $q > 0$ for all $x \in [-\pi, \pi]$, into a variational problem, stating the trial and test function spaces.




**BC1**: $u(-\pi) = u(\pi)$ (same value)
**BC2**: $u'(-\pi) = u'(\pi)$ (same slope)

Periodic conditions do not prescribe anything *at* an end — they only require the two ends to *match*. Equivalently the interval is closed into a circle, on which there is no boundary at all. That is why, in (35), the two endpoint contributions cancel one another instead of each vanishing separately.

### Solution

**Who is who.** Before starting, fix the vocabulary:

| symbol | role | what it is |
|---|---|---|
| $u$ | **trial function** | the unknown we are solving for — the solution |
| $v$ | **test function** | arbitrary; the identity must hold for *every* $v$ |
| $V$ | **trial and test space** | the admissible set, constructed in Step 3 |

Here the trial space and the test space turn out to be the *same* set $V$ (Step 3). Mechanically $v$ is a virtual displacement: "for all $v \in V$" means "for every admissible virtual motion".

**Step 1: multiply by a test function and integrate by parts.** Multiply (32) by $v$ and integrate over $(-\pi, \pi)$:

$$\int_{-\pi}^{\pi} (-u'')\,v\,\mathrm{d}x \ + \ \int_{-\pi}^{\pi} q\,u\,v\,\mathrm{d}x \ = \ \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x. \tag{33}$$

Only the first term needs work. Taking it one step at a time:

$$\begin{aligned}
\int_{-\pi}^{\pi} (-u'')\,v\,\mathrm{d}x &= -\int_{-\pi}^{\pi} u''\,v\,\mathrm{d}x && \text{pull the minus sign out} \\
&= -\Big[\,u'v\,\Big]_{-\pi}^{\pi} + \int_{-\pi}^{\pi} u'v'\,\mathrm{d}x && \text{integrate by parts: } (u'v)' = u''v + u'v' \\
&= -\Big(u'(\pi)v(\pi) - u'(-\pi)v(-\pi)\Big) + \int_{-\pi}^{\pi} u'v'\,\mathrm{d}x && \text{evaluate the bracket at both ends.}
\end{aligned}$$

Substituting back into (33),

$$\int_{-\pi}^{\pi} u'v'\,\mathrm{d}x + \int_{-\pi}^{\pi} q\,u\,v\,\mathrm{d}x \ \underbrace{-\Big(u'(\pi)v(\pi) - u'(-\pi)v(-\pi)\Big)}_{\text{boundary term}} \ = \ \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x. \tag{34}$$

**Step 2: the boundary term.** This is where **BC2** is used. Since $u'(\pi) = u'(-\pi)$, write $u'_\partial$ for that common value; it factors out:

$$-\Big(u'(\pi)v(\pi) - u'(-\pi)v(-\pi)\Big) = -\,u'_\partial\Big(v(\pi) - v(-\pi)\Big). \tag{35}$$

So the boundary term vanishes *if and only if the test function is itself periodic*, $v(-\pi) = v(\pi)$.

*Note.* The factor $u'_\partial$ is *not* zero. It is the slope of the solution at the seam, whatever value that turns out to be; we never learn it and never need it — it only has to be *finite*, so that finite$\times 0 = 0$. The product (35) vanishes because the *second* factor does. This is also why **BC2** alone cannot kill the boundary term: it only allows the two slopes to be combined into one common factor. Killing that factor then requires $v$ to be periodic, which is **BC1** imposed on the test function through $V$. Note too the mechanism: the two endpoint contributions *cancel each other*, rather than each vanishing separately as they did under homogeneous Dirichlet conditions.

**Step 3: the function space.** Periodicity must therefore be built in:

$$V := \Big\{\, v \;:\; \int_{-\pi}^{\pi} \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty, \quad v(-\pi) = v(\pi) \,\Big\}. \tag{36}$$

The **trial space and the test space are both** $V$. They coincide because $v(-\pi) - v(\pi) = 0$ is a *homogeneous linear* constraint, so $V$ is a vector space — unlike the nonhomogeneous Dirichlet case, where the trial set is only an affine shift of the test space.

**Step 4: the variational problem.** For $v \in V$ the boundary term now vanishes — but note it takes *two* facts, not one: by (35) it equals $-u'_\partial\big(v(\pi) - v(-\pi)\big)$, and $v(\pi) = v(-\pi)$ because $v \in V$ by (36). Equation (35) on its own is only a rewriting; it is the periodicity of $v$ that makes the factor zero. Dropping the term from (34):

$$\boxed{\text{find } u \in V \;\text{ such that }\; \int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x \;=\; \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \forall\, v \in V.} \tag{37}$$

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### Which condition is essential, which is natural

**BC1 is essential**: it is imposed in the definition (36) of $V$ and never appears in the equation.
**BC2 is natural**: it is never imposed, and it falls out of (37).
To verify the second claim, run the integration by parts *backwards*, starting from (37):

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \text{the variational problem (37)}$$

$$\int_{-\pi}^{\pi} u'v'\,\mathrm{d}x = \Big[\,u'v\,\Big]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} u''\,v\,\mathrm{d}x \qquad \text{integrate by parts, the other way}$$

$$\Longrightarrow \quad \Big[\,u'v\,\Big]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} u''v\,\mathrm{d}x + \int_{-\pi}^{\pi} q\,u\,v\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \text{substitute back}$$

Collecting everything on one side:

$$\int_{-\pi}^{\pi} \big(-u'' + q\,u - f\big)\,v\,\mathrm{d}x \;+\; \Big[\,u'v\,\Big]_{-\pi}^{\pi} \;=\; 0 \qquad \forall\, v \in V. \tag{38}$$

**Stage 1 — recover the ODE.** Restrict to those $v \in V$ that additionally vanish at both ends, $v(-\pi) = v(\pi) = 0$:

$$\begin{aligned}
& \Big[\, u'v \,\Big]_{-\pi}^{\pi} = 0 && \text{both endpoint factors } v(\pm\pi) \text{ vanish} \\
\Longrightarrow \quad & \int_{-\pi}^{\pi} \big(-u'' + q\,u - f\big) v \,\mathrm{d}x = 0 && \text{only the integral survives in (38)} \\
\Longrightarrow \quad & -u'' + q\,u = f \quad \text{on } (-\pi, \pi) && \text{fundamental lemma; the integrand is continuous}
\end{aligned}$$

**Stage 2 — release the endpoints, and BC2 falls out.** Now take a *general* $v \in V$:

$$\begin{aligned}
& \int_{-\pi}^{\pi} \big(-u'' + q\,u - f\big) v \,\mathrm{d}x = 0 && \text{the integrand vanishes, by Stage 1} \\
\Longrightarrow \quad & \Big[\, u'v \,\Big]_{-\pi}^{\pi} = 0 && \text{all that remains of (38)} \\
& u'(\pi)v(\pi) - u'(-\pi)v(-\pi) = 0 && \text{write the bracket out} \\
& \big(u'(\pi) - u'(-\pi)\big)\, v_\partial = 0 && v \in V \Rightarrow v(\pi) = v(-\pi) =: v_\partial \\
\Longrightarrow \quad & u'(\pi) = u'(-\pi) && v_\partial \text{ arbitrary; e.g. } v \equiv 1 \in V
\end{aligned} \tag{39}$$

which is **BC2**.

*Note.* Why $q > 0$ is given. Constants belong to $V$ — a constant is periodic. For the pure operator $-u''$ they would lie in its kernel and the problem would be singular, the periodic analogue of an unconstrained rigid-body mode. The term $q\,u$ removes it: putting $u = v = c$ in (37) makes the left-hand side $\int_{-\pi}^{\pi} q\,c^2 \,\mathrm{d}x > 0$ for any constant $c \neq 0$. So $q > 0$ is what makes the left-hand side of (37) positive definite here, doing the job the supports did in Problem 1.

## Problem 3(b)

**Problem.** Interpret the variational problem (37) as an energy minimization problem, stating the energy functional, and prove that the two are equivalent.

### The energy functional

For $w \in V$ define

$$F(w) := \tfrac{1}{2}\int_{-\pi}^{\pi}\Big((w')^2 + q\,w^2\Big)\,\mathrm{d}x \;-\; \int_{-\pi}^{\pi} f\,w\,\mathrm{d}x. \tag{40}$$

**Minimization problem:**

$$\text{find } u \in V \text{ such that } \;\; F(u) \le F(w) \quad \forall\, w \in V. \tag{41}$$

*Note.* Mechanically $F = U - W$: the first integral is stored energy — $(w')^2$ the strain part and $q\,w^2$ an elastic-foundation term — and the second is the work done by the load. So (41) is the principle of minimum total potential energy.

### The key expansion

Both directions rest on one computation: substitute $w = u + v$ into (40) and expand the two squares, using nothing but $(a+b)^2 = a^2 + 2ab + b^2$ (the factor 2 cancels the $\frac{1}{2}$, which is where the clean middle terms come from):

$$\begin{aligned}
\tfrac{1}{2}\int_{-\pi}^{\pi}(u'+v')^2\,\mathrm{d}x &= \tfrac{1}{2}\int_{-\pi}^{\pi}(u')^2\,\mathrm{d}x + \int_{-\pi}^{\pi}u'v'\,\mathrm{d}x + \tfrac{1}{2}\int_{-\pi}^{\pi}(v')^2\,\mathrm{d}x\\
\tfrac{1}{2}\int_{-\pi}^{\pi}q\,(u+v)^2\,\mathrm{d}x &= \tfrac{1}{2}\int_{-\pi}^{\pi}q\,u^2\,\mathrm{d}x + \int_{-\pi}^{\pi}q\,u\,v\,\mathrm{d}x + \tfrac{1}{2}\int_{-\pi}^{\pi}q\,v^2\,\mathrm{d}x\\
-\int_{-\pi}^{\pi}f\,(u+v)\,\mathrm{d}x &= -\int_{-\pi}^{\pi}f\,u\,\mathrm{d}x \;-\; \int_{-\pi}^{\pi}f\,v\,\mathrm{d}x
\end{aligned}$$

Now sort the pieces by *how many $v$'s each contains*:

| | collected terms | what it is |
|---|---|---|
| no $v$ | $\frac{1}{2}\int(u')^2\,\mathrm{d}x + \frac{1}{2}\int q\,u^2\,\mathrm{d}x - \int f\,u\,\mathrm{d}x$ | $F(u)$ |
| one $v$ | $\int u'v'\,\mathrm{d}x + \int q\,u\,v\,\mathrm{d}x - \int f\,v\,\mathrm{d}x$ | the residual of (37) |
| two $v$'s | $\frac{1}{2}\int(v')^2\,\mathrm{d}x + \frac{1}{2}\int q\,v^2\,\mathrm{d}x$ | $\frac{1}{2}\int\big((v')^2 + q\,v^2\big)\,\mathrm{d}x$ |

Adding the three rows:

$$F(u+v) = F(u) \;+\; \underbrace{\left[\int_{-\pi}^{\pi}\big(u'v' + q\,u\,v\big)\,\mathrm{d}x - \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x\right]}_{\text{residual of (37)}} \;+\; \tfrac{1}{2}\int_{-\pi}^{\pi}\Big((v')^2 + q\,v^2\Big)\,\mathrm{d}x. \tag{42}$$

The middle bracket is linear in $v$; the last term is quadratic. The scalar analogue is a single spring, $F(x) = \frac{1}{2}kx^2 - Px$, for which $F(x+y) = \big[\frac{1}{2}kx^2 - Px\big] + y\,(kx - P) + \frac{1}{2}ky^2$ — the same three pieces: a constant, a term linear in the perturbation carrying the residual, and a non-negative quadratic. These three pieces do different jobs, and that is the whole proof: the *linear* term vanishes exactly when $u$ solves (37), while the *quadratic* term is $\ge 0$ always. Direction (i) kills the linear term and reads off $F(w) \ge F(u)$; direction (ii) reads the same linear term off as $g'(0)$.

### (i) Variational $\Longrightarrow$ minimization

Let $u$ solve (37). Given any $w \in V$, set $v := w - u$. Then $v \in V$, *because $V$ is a vector space* — the periodicity constraint is homogeneous, so a difference of two admissible functions is admissible. Applying (42), the middle bracket vanishes by (37), leaving

$$F(w) = F(u) + \tfrac{1}{2}\int_{-\pi}^{\pi}\Big((v')^2 + q\,v^2\Big)\,\mathrm{d}x \;\geq\; F(u), \tag{43}$$

since $q > 0$ makes the integrand non-negative. Hence $u$ solves (41). ■

### (ii) Minimization $\Longrightarrow$ variational

Let $u$ solve (41) and fix an arbitrary $v \in V$. Since $V$ is a vector space, $u + \epsilon v \in V$ for every $\epsilon \in \mathbb{R}$, so by (42) with $v$ replaced by $\epsilon v$,

$$g(\epsilon) := F(u+\epsilon v) = F(u) + \epsilon\left[\int_{-\pi}^{\pi}\big(u'v' + q\,u\,v\big)\,\mathrm{d}x - \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x\right] + \tfrac{1}{2}\epsilon^2\int_{-\pi}^{\pi}\Big((v')^2 + q\,v^2\Big)\,\mathrm{d}x. \tag{44}$$

This is a quadratic polynomial in $\epsilon$ with a positive coefficient of $\epsilon^2$. Because $u$ minimises $F$, the function $g$ attains its minimum at $\epsilon = 0$, so $g'(0) = 0$. Differentiating (44),

$$g'(0) = \int_{-\pi}^{\pi}\big(u'v' + q\,u\,v\big)\,\mathrm{d}x - \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x = 0, \tag{45}$$

which is (37). As $v \in V$ was arbitrary, $u$ solves the variational problem. ■

*Note.* Where $q > 0$ earns its keep. For the inequality in (43) only $(v')^2 \geq 0$ is needed. Strictness needs $q > 0$: $\int\big((v')^2 + q\,v^2\big)\,\mathrm{d}x = 0$ forces $v' \equiv 0$ *and* $v \equiv 0$. Without the $q\,v^2$ term a constant $v$ would give zero — and constants do belong to $V$. So $q > 0$ is what makes the minimiser *unique*, and what makes the $\epsilon^2$ coefficient in (44) strictly positive, so that $\epsilon = 0$ is a genuine minimum rather than merely a stationary point.